# Behind the Lens:
# Making Home CCTV Data Practices Visible

## A1.1 Problem Identification

ANN

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## 1. Introduction

Home CCTV and smart security cameras are increasingly used for home security, pet monitoring and family care. Although these systems are designed to provide safety and convenience, their operation can involve local storage, cloud services, remote access, provider infrastructure and different forms of data access that are not always visible to users. This project investigates how home CCTV users understand what happens to their video data, with particular attention to data transmission, storage, retention and access.

Our initial research combines background literature, government guidance, documented privacy and security incidents, a comparison of commonly available CCTV systems, and interviews with five participants representing current users and a potential buyer. The research indicates that people can be comfortable using CCTV applications while remaining uncertain about how footage travels through the wider system or whether manufacturers and other parties may be able to access it. Participants also highlighted permission, transparency and clear privacy information as important factors in trust.

Based on these findings, the report identifies a design opportunity to make CCTV data practices more visible and understandable. It explores several provisional concepts, including a privacy label, an interactive privacy journey, a contextual privacy dashboard, and access history with notifications. The report also considers technical obstacles, ethical issues and project risks that will guide the next stage of prototyping and evaluation.

## 2. Problem Overview & Background Research

### 2.1 Problem Observation and Initial Problem Statement

Home security cameras are becoming increasingly common in everyday households. People may use them to monitor entrances and outdoor areas, protect their property, or keep an eye on children and pets when they are away from home. These devices are generally purchased with the expectation that they will improve safety and provide users with greater peace of mind.

However, reports of privacy and security incidents involving connected security cameras have raised concerns about the way these devices handle personal video data. This led our team to question what happens to CCTV footage after it is recorded. For example, is the footage stored locally on the device or uploaded to the cloud? How is it transmitted when users access their cameras remotely? Who is able to access the data, and under what circumstances could a service provider or another party gain access to it? We also became interested in how long recorded data is retained and what protections are used to secure it.

At this stage, we do not assume that CCTV providers can freely access users' recordings or that all products handle data in the same way. Instead, our concern is that these processes may differ significantly between products and may not always be clear to ordinary consumers.

This concern is supported by previous research involving users of smart cameras and video doorbells. Tabassum et al.(Tabassum et al., n.d.) conducted interviews with 23 smart-home users to investigate how they understood the collection, storage, sharing and use of data generated by connected devices. Their study included users of smart cameras and video doorbells and identified differences in users' awareness of video storage practices. Participants who had personally installed their camera or video doorbell were generally aware of video deletion options or how long recordings would remain on the provider's server. In contrast, participants who had not been involved in installing these devices were uncertain about video storage policies and available deletion options. The study also found that users could be aware of the primary data collected by a camera, such as video recordings, while overlooking other data collected through associated applications.

These findings do not demonstrate that all home CCTV users lack an understanding of their data. Instead, they suggest that users' awareness of camera data practices may vary depending on their involvement with the device and their exposure to information about how the service operates. This is particularly relevant to our project because ordinary household members may use a security camera without having selected, installed or configured it themselves. It therefore remains important to investigate whether consumers clearly understand what happens to their camera data beyond the visible act of recording video.

This leads to our initial problem statement:

**Users may not clearly understand how their home CCTV data is transmitted, stored and accessed, or who may have access to it.**

## 2.2 Validation of the problem

The Office of the Australian Information Commissioner (OAIC)has recognised that Internet of Things (IoT) devices can create significant privacy and security risks because they are increasingly interconnected and capable of collecting large amounts of personal and sometimes sensitive information. The OAIC has also highlighted problems with how these data practices are communicated to consumers. In a global privacy sweep of IoT products, 71% of the devices and services examined failed to properly explain how information was stored, 69% did not adequately explain how users could delete their information, and 91% did not advise users to customise their privacy settings(OAIC, 2023).

Although these findings relate to IoT products more broadly rather than home security cameras specifically, they are relevant to our project because modern home CCTV systems are internet-connected devices that can involve mobile applications, remote-access services, cloud infrastructure and

data storage. Video footage captured in and around a home may also contain highly personal information about household members, visitors and everyday activities. Therefore, understanding how this information is collected, transmitted, stored and accessed is particularly important in the context of home surveillance.

The OAIC has also discussed the challenges of relying on notice and consent to help individuals manage their personal information. For consent to be meaningful, users need to be able to understand how their information is being collected, used and disclosed. However, IoT systems can be technically complex, making these practices difficult for ordinary consumers to understand. This suggests that simply providing a privacy policy or notice does not necessarily mean that users will develop a clear understanding of what happens to their data.

Evidence of this issue can also be found in research involving users of smart cameras and video doorbells.(Tabassum et al., n.d.)investigated users' understanding of data collection, sharing, storage and use across connected household devices. Their study found a noticeable difference between users who had personally installed cameras or video doorbells and those who had not. Participants who installed these devices themselves were generally aware of video deletion options or how long recordings would remain on a provider's server. In contrast, participants who had not been involved in installation were uncertain about video storage policies and available deletion options. The researchers also found that camera and video-doorbell users were aware that their devices collected video footage, but could overlook other information collected through the applications associated with those devices.

These findings are particularly relevant to our project because a home security camera may be used by several members of a household even though only one person selected, installed or configured the device. As a result, access to the camera does not necessarily mean that every user has been exposed to information about its storage, deletion or other data-handling practices. The findings therefore provide evidence that understanding of camera data practices can vary between users and that some users may have limited awareness of how their recorded data is managed.

The Australian Government has also introduced mandatory baseline cybersecurity requirements for consumer smart devices. The Cyber Security (Security Standards for Smart Devices) Rules 2025 commenced on 4 March 2026 following a 12-month transition period. For most consumer smart devices manufactured from this date, the Rules require measures including the removal of universal default passwords, a published mechanism for reporting security issues, and information about how long manufacturers will provide security updates(*Department of Home Affairs Website*, n.d.).

The introduction of these requirements demonstrates that the security of connected consumer devices has become an issue formally recognised and regulated by the Australian Government. However, these requirements mainly address baseline cybersecurity practices. They do not themselves explain to consumers where home CCTV footage is stored, how footage is transmitted when cameras are accessed remotely, how long recordings are retained, or under what circumstances a service provider or another party may be able to access the data.

Taken together, the government and academic evidence supports our concern that privacy, transparency and user understanding are relevant issues in connected home-camera environments. However, the available evidence does not establish that all CCTV users lack an understanding of their data, nor does it establish how well Australian home CCTV users understand these processes specifically. This remains

an important gap in our current understanding and provides a direction for further investigation into how users perceive the storage, transmission, retention and access of their home CCTV data.

## 2.3 Real-world CCTV Privacy and Security Incidents

The previous section established that transparency and user understanding are relevant concerns in connected home-camera systems. To further investigate the practical significance of these concerns, we reviewed several publicly documented privacy and security incidents involving connected security cameras. Rather than treating these incidents simply as examples of cameras being “hacked”, we examined which parts of the wider camera system were involved and what they reveal about the handling and access of users' video data.

Several incidents demonstrate that privacy failures can occur at different points within a connected camera system. In 2021, Eufy experienced a software error during a server update that affected users in several countries, including Australia, with some users reporting that they were able to view camera feeds and account information belonging to other households.(*Updated Official Statement - Archived*, 2021) Although Eufy stated that recorded video was primarily stored locally on users' devices, services such as account management, device management and remote P2P access still relied on online server infrastructure. This case shows that local storage does not necessarily mean that a camera operates independently of external services, particularly when remote access and account management are involved.

A different type of privacy issue was identified in the case of Ring. The U.S. Federal Trade Commission alleged that Ring had failed to adequately restrict employee and contractor access to customers' private videos and had also failed to implement sufficient protections against unauthorised access to user accounts. Unlike the Eufy incident, this case was not primarily concerned with where footage was stored. Instead, it highlighted the importance of provider-side access controls and raised the question of who within a service provider may be technically or organisationally able to access customer recordings.

A further example occurred in 2024, when Wyze reported a backend security incident following a service outage.(*Update on Investigation of 2/16/24 Security Issue - Wyze News*, 2024) Approximately 13,000 users were affected, with some receiving camera event thumbnails that belonged to other users. Wyze attributed the incident to a third-party caching library that incorrectly associated device IDs with user IDs while systems were recovering from the outage. This case demonstrates that privacy failures can also occur within backend infrastructure even when users' accounts and physical cameras themselves have not been directly compromised.

Although the three incidents resulted in similar privacy concerns, the underlying causes were different. The Eufy case highlighted the role of remote-access and server infrastructure in a system that primarily promoted local storage; the Ring case demonstrated the importance of provider-side human access and access-control policies; and the Wyze incident showed how backend systems can incorrectly associate camera data with user accounts.

Together, these cases changed our initial understanding of the problem. Connected-camera privacy cannot be evaluated solely by asking whether footage is stored locally or in the cloud, nor can privacy risks be understood only in terms of whether the physical camera can be hacked. The wider system may also involve account management, remote-access services, backend infrastructure, third-party components and provider-side access controls, each of which can influence how video data is handled and who may be able to access it.

This finding is important for our project because many of these processes are largely invisible during ordinary camera use. It also suggests that comparing home security cameras simply as “local” or “cloud” products would provide an incomplete picture of their privacy characteristics. Therefore, our next stage of investigation considers a broader range of factors across commonly available camera systems, including storage methods, cloud involvement, remote access and P2P services, encryption, privacy policies and conditions of data access.

## 2.4 Key Findings from the Comparison

| Brand | Storge Approach | Cloud/Remote Access | Encryption | Data Access/Privacy |
|---|---|---|---|---|
| Eufy | Mainly local via device/HomeBase; optional cloud | Remote app access still involves online services | Encryption used for stored/transmitted data; E2EE available in some configurations | Access mainly through the user's account/shared users; provider access depends on service and configuration |
| Ring | Mainly cloud-based recording | Strong reliance on Ring cloud and app for remote access | Encryption used; optional E2EE on compatible devices | Account owner/shared users can access recordings; provider-side access is governed by access controls and privacy policies |
| Tapo | microSD/local storage + optional Tapo Care cloud | Remote viewing through Tapo app; cloud recording optional | Encryption used for data transmission; varies by model | Access through authorised Tapo accounts; cloud data handled under TP-Link/Tapo policies |
| Reolink | Mainly local via microSD, Home Hub or NVR; optional cloud | Remote access supported, including UID/P2P on supported devices | Encryption varies by device/service | Device/account owner and authorised shared users; cloud involvement depends on configuration |
| Arlo | Cloud recording + optional local storage through compatible SmartHub | Remote access primarily through Arlo app/cloud services | Encryption used for transmission and stored cloud video | Account owner and authorised users can access recordings; handling depends on cloud/local configuration |

The comparison shows that home security cameras offering similar functions can rely on substantially different approaches to storing and accessing video data. Ring relies heavily on cloud-based services, while Eufy and Reolink place greater emphasis on local storage, and Tapo and Arlo provide combinations of local and cloud options depending on the device and configuration. However, the distinction between “local” and “cloud” is not always straightforward. A camera that stores recordings locally may still rely on online services for account management, notifications or remote viewing.

Therefore, the comparison of commonly available home security camera brands suggests that simply knowing whether recordings are stored locally or in the cloud is not sufficient to fully understand how a camera handles user data. Consumers may also need to understand relatively technical aspects, such

as how remote access is provided, which online services are involved, what protections are applied to the data, and who is authorised to access it. For ordinary consumers, these differences may not be easy to identify or understand when comparing products in stores or on websites. This further suggests that the user-understanding problem identified in our project may exist in practice and warrants further investigation.

# 3. Stakeholders

## 3.1 Stakeholder Identification

Our project involves different stakeholders who may use, interact with, be recorded by, or influence a home CCTV system. We identified the stakeholders based on their relationship with CCTV data, privacy, access and control.

### *3.1.1 Primary Stakeholders*

**CCTV Owners / Users:** CCTV owners are the main users of the system. They use cameras and mobile apps to view recordings, monitor their homes and manage CCTV settings. Our research focuses particularly on how these users understand where their CCTV data is stored, how it is transmitted, and who may have access to it.

**Household Members:** Family members or other people living in the home may be recorded regularly, even if they are not the person who purchased or manages the CCTV system. Their privacy and awareness should therefore also be considered.

**Potential CCTV Buyers:** People considering buying CCTV are important because privacy, storage and data-access information may affect their purchasing decisions. This stakeholder group is represented by P3 in our user research.

### *3.1.2 Secondary Stakeholders*

**Visitors:** Friends, relatives, delivery workers or other visitors may be captured by home CCTV without directly interacting with the system. They may have limited knowledge or control over how their images are recorded or stored.

**Neighbours:** Depending on the position and viewing angle of a camera, neighbouring properties or people may also appear in CCTV footage. This creates privacy considerations beyond the CCTV owner's household.

### *3.1.3 System and Service Stakeholders*

**CCTV Manufacturers:** Manufacturers provide the camera hardware, software and associated services. Depending on the system, they may also provide account management, remote-access functions and other infrastructure.

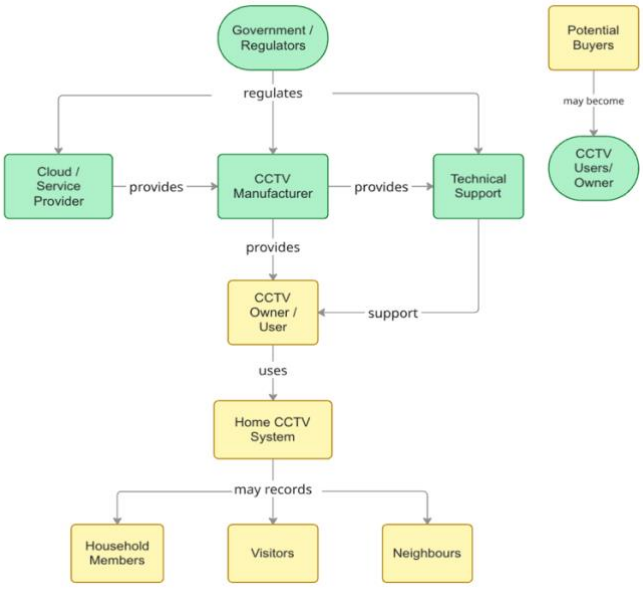
**Cloud / Service Providers:** Some CCTV systems use cloud services for video storage, processing or remote access. These services form part of the wider data flow between the camera and the user's device.

**Technical Support:** Technical-support staff may help users manage CCTV accounts, devices and system problems. Their possible relationship with account or system access makes them relevant to the project's privacy considerations.

### *3.1.4 External Stakeholders*

**Government / Regulators:** Government and regulatory bodies influence privacy, data protection and security requirements. They are indirect stakeholders because their policies and guidance can influence how CCTV companies manage and communicate user data practices.

## 3.2 Stakeholder Map




The stakeholder map places the CCTV owner at the centre of the user experience while showing that CCTV privacy involves more than the owner alone. Household members, visitors and neighbours may be recorded, while manufacturers, cloud providers and technical support may form part of the wider technical system. Potential buyers are also relevant because privacy information may influence decisions before CCTV adoption.

# 4. User Research

## 4.1 Participants

| Participant | Background | CCTV Experience | Main Use / Motivation |
|---|---|---|---|
| **P1** | 26, female, lives with family and has a dog | Current user, 2 cameras | Monitoring a dog and an elderly family member |
| **P2** | 21, female, lives with family and has a dog | Current user, 3 cameras | Monitoring a dog and elderly family members |
| **P3** | 28, lives alone, full-time worker | Potential buyer, has never owned CCTV | Considering CCTV for home security and theft prevention |
| **P4** | 52, female, mother, lives with husband and 19-year-old child | Long-term user, around 7 years | Originally installed CCTV to check on her child while parents were at work |
| **P5** | Lives alone and has a cat | Current smart CCTV user | Pet monitoring and home security |

Five participants with different CCTV experiences and household situations were included. Four participants were current or long-term CCTV users, while P3 was a potential buyer who had never owned CCTV. This provided perspectives from both existing use and pre-purchase decision-making.

## 4.2 Semi-structured Interviews

We conducted semi-structured interviews with five participants to explore how current and potential CCTV users understand privacy, data storage, data access and control. Four participants had experience using home CCTV, while one participant was considering purchasing a CCTV system.

Each interview lasted approximately 10 minutes. We used six main questions with follow-up questions, allowing participants to explain their personal experiences and concerns in more detail. The interviews focused on:

- participants' current or expected use of CCTV;
- their understanding of where CCTV recordings are stored;
- who they believe can access their recordings;
- their awareness of manufacturer or third-party access;
- their experience with privacy information and policies; and
- what information or controls would help them better understand and trust their CCTV system.

Participants were encouraged to answer based on their own experience and understanding. The interview data was later analysed using Affinity Mapping to identify common patterns, differences and key findings.

## 4.3 Mental Model Activity

The Mental Model Activity asked participants to explain or draw how they thought CCTV video travels from the home camera to the phone app.

P1 — Understands the basic data flow
P1 thought the video goes from the home camera, through Wi-Fi and the cloud, and then to the mobile app.

> P1: "I think the camera connects to our home Wi-Fi first. Then the video is uploaded through Wi-Fi to the cloud used by the app."

***Mental Model: Home Camera → Home Wi-Fi → App / Cloud → Mobile App***

P2 — Unclear about the data flow

P2 knew how to use the app, but did not understand how the video gets from the camera to the phone.

> P2: "I'm not sure. This is quite difficult."

> P2: "I know that I can open the app on my phone and watch the camera, but I don't really know how the video actually gets from the camera to the app."

***Mental Model: Camera → Wi-Fi → Phone App (?)***

P3 — Thinks the manufacturer cloud is involved

P3 thought the video probably goes through Wi-Fi, the internet and the CCTV company's cloud. However, she was unsure what the cloud actually does.

> P3: “Then it probably sends the video through the internet somewhere... maybe to the company's cloud?”
>
> P3**:** “But I'm not sure whether the company actually stores the video or whether it just passes through their system.”

***Mental Model: Camera → Home Wi-Fi → Internet → Cloud / CCTV Company? → Phone App***

P4 — No Mental Model Activity recorded

The document does not contain a completed Mental Model Activity for P4. Therefore, we should not assume P4's understanding of the data flow.

P5 — Understands part of the system, but is still uncertain

P5 thought the video goes through Wi-Fi and then through the manufacturer's system or server before reaching the phone.

> P5: “I think the camera first connects through the network or Wi-Fi.”
>
> P5: “Then the video probably goes through the manufacturer's system or server.”
>
> P5**:** “But I'm not completely sure what happens in the middle.”

Mental Model Summary
The participants showed different levels of understanding. They generally understood the connection between the camera and phone app, but some were unsure about what happens between them, especially the role of the cloud and manufacturer servers.

## 4.4 Affinity Mapping

We grouped similar comments from P1–P5 into five themes.

### *4.4.1 Theme 1 — Understanding Data Storage and Flow*

Several participants were unsure about where CCTV recordings are stored or how the data travels.

> P1: “I'm not completely sure... I think the internet or cloud might also be involved.”
>
> P2: “I think they're probably stored in the cloud.” / “I'm not completely sure.”
>
> P3: “I guess probably in the cloud?”
>
> P5: “I have a cloud storage subscription, so I think most of my recordings are stored on the manufacturer's cloud servers.”

Pattern: Users have different levels of understanding about CCTV storage and data flow.

### *4.4.2 Theme 2 — Manufacturer Access and User Control*

> P1: “I'm not sure whether they technically can, but I definitely wouldn't want them to access my recordings without my permission.”
>
> P2: “I think I have the right to know if someone accesses my CCTV recordings.”
>
> P3: “I'd want to know who accessed it, when they accessed it and why.”

> P4: “I'm actually not sure whether they technically can. I just don't think they should be allowed to access private recordings from inside our home.”

Pattern: Users want more control and awareness of who can access their CCTV data.

### *4.4.3 Theme 3 — Privacy and Trust*

Unauthorised access and data leaks were strong concerns.

> P1: “If the company stole or accessed my recordings without my permission... I would completely lose trust in them.”
> P2: “It would be even worse if our private recordings were leaked or uploaded online.”
> P4: “If the company accessed or took my recordings without my permission, I would lose trust in them.”
> P5: “The biggest thing would be a data leak, especially if users' private camera footage was leaked.”

Pattern: Privacy problems can directly reduce trust in CCTV companies.

### *4.4.4 Theme 4 — Privacy Information*

Participants interacted with privacy information differently.

> P1: “No, I didn't really read it.”

> P2: “We didn't really pay much attention to the privacy information.”

> P4: “No, I didn't read it myself. My husband read the information when we set up the camera.”

P5 was different because she had read the privacy information and understood that cloud recordings were uploaded to manufacturer servers.

Pattern: Not all users actively read or understand privacy information.

### *4.4.5 Theme 5 — Need for Clear Transparency*

Several participants wanted clearer information about their CCTV data.

> P1: “I would feel safer if the manufacturer clearly told us what happens to our recordings.”

> P2: “I would mainly like to know more clearly who can access the CCTV recordings from my home.”

> P3: “The most important thing for me is probably who can see the video. I also want to know where it's stored.”

> P5: “I don't want a very long privacy document.” She preferred “a short pop-up with simple bullet points.”

However, P4 said:

> P4: “Not really at the moment. I haven't really thought much about these questions before.”

Pattern: Many users want clearer privacy information, but not every user actively looks for it.

## 4.5 Key Findings

### *4.5.1 Finding 1 — Users can use CCTV without fully understanding the data flow.*

Several participants knew how to watch CCTV through an app, but they were less certain about what happens to the video between the camera and phone. P2 said she did not know "how the video actually gets from the camera to the app," while P5 was "not completely sure what happens in the middle."

### *4.5.2 Finding 2 — Manufacturer access is unclear to users.*

Participants were unsure whether CCTV companies could technically access their recordings. At the same time, they wanted permission or notification when access happens. P1 said she did not want access "without my permission," and P2 said she had "the right to know."

### *4.5.3 Finding 3 — Privacy problems can damage trust.*

Participants said unauthorised access and data leaks would reduce their trust in CCTV companies. P4 said she would lose trust if a company accessed her recordings without permission, while P5 identified data leaks as a major concern.

### *4.5.4 Finding 4 — Privacy information is not always read.*

P1 did not read the privacy information, P2 paid little attention to it, and P4 relied on her husband during setup. This shows that privacy information is not always actively read by the person using the CCTV.

### *4.5.5 Finding 5 — Several users want clearer privacy information, but this need is not always actively recognised.*

Participants showed different levels of CCTV privacy awareness. P1, P2, P3, and P5 wanted clearer information about data storage, access, and handling, while P4 had rarely considered these issues.

This suggests that users may not think about privacy until they are asked, experience a problem, hear about a data leak, or consider buying a device. Therefore, low concern does not necessarily mean low privacy needs.

P3 also provides a pre-purchase perspective, while the other participants mainly reflect experiences after CCTV adoption.

### *4.5.6 Overall Research Insight*

Users know how to use CCTV, but they do not always understand what happens to their video data. The biggest uncertainty is around data flow and manufacturer access. Participants also showed that privacy, permission and clear information are important for trust.

# 5. Design Opportunity and Initial Conceptual Model

## 5.1 Existing Solutions

Current home CCTV systems already provide different privacy and security functions to help users protect and manage their camera data. Common solutions include privacy policies, app privacy settings, login and security alerts, shared-user management, and cloud or local storage information. These functions provide users with some information and control, but they usually focus on different parts of the CCTV system.

For example, Ring provides a Control Centre where users can manage authorised devices, shared users, linked accounts, video storage and video encryption(Ring Control Centre Information, n.d.). Ring also provides optional end-to-end encryption (E2EE) for compatible devices, which gives users additional protection for their video recordings(*Using Video End-to-End Encryption (E2EE)*, n.d.). Shared-user

management also allows camera owners to give family members or other trusted people access to selected devices(OAIC, 2023).

Privacy policies are another common solution. They explain how companies collect, store, use and share personal information. However, providing privacy information does not always mean that users can easily understand it. The Office of the Australian Information Commissioner (OAIC) reported that 71% of the IoT devices and services examined did not properly explain how information was stored, while 69% did not adequately explain how users could delete their information(*Managing Your Ring Account and Privacy Settings*, n.d.).

Current CCTV systems also provide security alerts and storage information. Security alerts can inform users about new devices or unusual account activity, while cloud and local storage information can help users understand where recordings are mainly stored. However, these solutions normally provide information separately rather than showing users a complete picture of how their CCTV data is handled.

| Existing Solution | Main Purpose | Current Limitation |
|---|---|---|
| Privacy Policy | Explains data collection, storage and sharing | Often long or complex, and users may not read it |
| App Privacy Settings | Allows users to manage privacy and security | Settings may be separated and use technical terms |
| Login / Security Alerts | Warns users about new logins or unusual activity | Mainly focuses on account activity rather than specific video access |
| Shared-user Management | Manages family members or other authorised users | Mainly shows users intentionally authorised by the owner |
| Cloud / Local Storage Information | Explains how recordings are stored | Does not always explain the complete data flow |

## 5.2 Limitations and Design Gap

Although current CCTV systems provide different privacy and security features, there are still limitations in user understanding and transparency. Our initial user research shows that some users can successfully use CCTV systems without fully understanding how their video data is handled.

One limitation is that privacy information is not always actively read or understood. In our interviews, P1 did not read the privacy information, P2 paid little attention to it, and P4 relied on her husband to review the information during setup. P5 was different because she had read the privacy information and had a better understanding of cloud storage. These results show that users have different levels of privacy awareness. This is also related to the OAIC's findings that many IoT products did not clearly explain data storage and deletion(*Managing Your Ring Account and Privacy Settings*, n.d.). Therefore, the problem is not only whether privacy information is available, but whether ordinary users can easily find and understand it.

Our Mental Model Activity also showed that users may not fully understand how CCTV data travels through the system. P2 knew that she could watch the camera through the mobile app but did not know how the video actually reached the app. P3 believed that the manufacturer's cloud was involved but was unsure whether the video was stored there or only passed through the system. Similarly, P5 thought that the video probably went through the manufacturer's server but was “not completely sure what happens in the middle.” This suggests a gap between using a CCTV system and understanding how the system handles data.

Another limitation is that current access-management tools do not always provide a complete picture of who may access CCTV data. Shared-user functions can show family members or other people intentionally authorised by the owner, while security alerts can show new account logins. However, several participants were particularly uncertain about manufacturer access. P1 said that she would not want a company to access her recordings without permission. P2 believed that she had "the right to know" if someone accessed her CCTV recordings, while P3 wanted to know who accessed the footage, when they accessed it and why. A historical case involving Ring also shows why provider-side access is important. In 2023, the U.S. Federal Trade Commission reported historical problems with Ring's employee and contractor access controls over customer videos(*FTC Says Ring Employees Illegally Surveilled Customers, Failed to Stop Hackers from Taking Control of Users' Cameras*, 2023). This case does not mean that CCTV companies can freely access all customer recordings today, but it shows why clear access information is important.

Overall, the main problem is not that current CCTV systems have no privacy or security solutions. Instead, privacy information and controls are often fragmented across privacy policies, settings, shared-user management, security alerts and storage information. Users may need to connect all of this information by themselves to understand what happens to their CCTV data.

Our research therefore shows a gap between:

> Providing privacy information and Helping users understand privacy information.

This creates a design opportunity to provide a simpler and more integrated way for users to understand where their CCTV data goes, where it is stored, who may access it, when access happens, and what they can control.

Based on this design gap, our current design question is:

> How might we help home CCTV users better understand and control who can access their camera data?

## 5.3 Initial Conceptual Model

At this stage, our conceptual model is intentionally high-level and exploratory. It does not represent a final product. Instead, it illustrates how a transparency layer could be added between the technical CCTV system and the user to make invisible data practices more understandable. The model will be used to communicate our current design ideas, compare alternative concepts and gather feedback before selecting a final direction.

Home CCTV Camera
↓
Video and related data are generated
↓
Data transmission
Wi-Fi / Internet / P2P / Provider services
↓
Data storage
Local / Cloud / Hybrid
↓
Different parties may have different access permissions
↓
OUR TRANSPARENCY LAYER
↓
Privacy Risk Label + Interactive Privacy Journey

+ Privacy Dashboard + Access History and Notifications

↓

USER

Understand → Compare → Decide → Control

### *5.3.1 Concept 1: Multi-level CCTV Privacy Label*

Existing privacy policies and product descriptions provide information about data collection and security, but this information is often long, technical and difficult to compare across products. To address this limitation, we propose a standardised, colour-coded privacy label that presents the most important privacy information before purchase and during device setup. The first layer of the label would provide an immediately visible privacy-exposure level using colour, icons and a short explanation. For example:

| **Level** | **Meaning** |
|---|---|
| Green | Lower privacy exposure |
| Light green | Relatively low exposure |
| Yellow | Moderate exposure |
| Orange | Relatively high exposure |
| Red | Higher privacy exposure |
| Grey | Insufficient verified information |

The rating could be based on clearly stated criteria rather than an unsupported general judgement. These criteria could include:

-Whether footage is stored locally, in the cloud or through a hybrid system

-Whether remote access requires the manufacturer's online services

-Whether end-to-end encryption is available and enabled by default

-How long recordings are retained

-Whether users can delete or export their recordings

-Whether third parties are involved in processing the data

-Whether the provider explains employee or technical-support access

-Whether access events are recorded and visible to users

The second layer would allow users to expand the label or scan a code to view detailed evidence, explanations and information sources. This layered design attempts to balance simplicity with transparency: non-technical users can quickly understand and compare the overall result, while users who want more information can access the detailed explanation. This concept responds to the limitations of existing privacy policies, which are difficult to read, and existing storage labels, which may oversimplify a product as simply "local" or "cloud." Research comparing four IoT security labels found that participants initially believed information-dense labels would be more useful, but after comparing products they tended to prefer designs that created less cognitive demand. This supports a layered approach combining a simple first-level rating with optional details

However, the label should not claim that a product will or will not experience a data leak. It communicates relative privacy exposure based on available evidence, not a prediction of future security incidents. The grey "insufficient information" category is also important because the design should not give an apparently precise rating when the manufacturer has not disclosed enough information.

### 5.3.2 Concept 2: Static and Dynamic Interactive Privacy Journey

Traditional privacy agreements are primarily static documents. Although they may contain legally complete information, users must read long sections of text and translate technical descriptions into an understanding of what happens during actual camera use.

Our second concept retains the full privacy agreement as a static and legally accessible document, but adds a short interactive privacy journey during device setup. Rather than asking users to immediately accept a long policy, the system would explain the most important stages through a simple visual flow:

Camera captures video

- Video travels through the home network

-Provider services may be involved

-Video is stored locally or in the cloud

- Authorised parties may access it

- The recording is retained or deleted

At each stage, the user could select "Why?", "Who can access this?" or "Can I change this?" to receive a short plain-language explanation. Important decisions would be presented contextually. For example, when cloud storage is selected, the interface would immediately explain where the recording is stored, the retention period and how the user can delete it.

The process could conclude with a short personalised summary:

Your current setup uses cloud storage. Recordings are retained for 30 days. Three household accounts have access. Remote viewing uses the provider's online service.

This concept responds to two limitations of current solutions. First, it reduces the cognitive burden created by long, static privacy policies. Second, it connects information to the user's actual configuration, rather than providing one general policy for every device and service combination.

The interaction must not replace meaningful consent with decorative animation. Users should still be able to pause, revisit information, access the full policy and change their choices before confirming the setup.

### 5.3.3 Concept 3: Contextual Privacy Dashboard

Existing control centres usually focus on account settings, shared users and individual security features. Our proposed dashboard would bring together the privacy information that is currently fragmented across policies, settings, subscription pages and support documents.

The dashboard could include:

-Current storage location and retention period
-Whether remote viewing uses provider infrastructure
-Current household users and their permissions
-Connected third-party services
-Encryption status with a plain-language explanation

-Options to delete, export or change the storage of recordings
-A “What has changed?” section after system or policy updates

Instead of only showing whether a function is turned on or off, the dashboard should explain the consequence of each setting. For example: Cloud storage is ON. This allows remote access and stores event recordings for 30 days.

This addresses the fragmentation of existing solutions and supports the user’s transition from simply seeing settings to understanding and controlling their consequences.

### *5.3.4 Concept 4: Access History with Contextual Notifications*

Existing security alerts usually focus on logins, device activity or motion events. They do not always show whether a particular recording was accessed, by whom, when or for what stated reason.

The proposed Access History would present different types of access separately:

-Household user viewed live video
-Shared user played a stored recording
-Connected service processed an event
-Technical-support access was requested and authorised
-Unusual device or account access was detected

Each entry could show:

Who or what accessed the data
Date and time
Type of access
Stated reason
Whether the access was authorised
What action the user can take

The system must clearly distinguish between three states:

1. A party has technical capability or permission to access data.
2. An automated service processed data.
3. A person actually viewed or accessed a recording.

This distinction is necessary because a message such as “Manufacturer has access” could be interpreted as “The manufacturer is currently watching me.” Notifications should therefore use factual and non-alarming language. Only unusual or high-risk events should trigger immediate alerts; routine system processing should remain visible in the history without creating unnecessary notification fatigue.

### *5.3.5 Current Conceptual Direction*

At this stage, these concepts should not be treated as four final product features. They represent alternative and potentially connected approaches that require further evaluation.

The current model suggests two main stages:

Before purchase: a multi-level privacy label helps users compare CCTV products.

During setup and everyday use: an interactive privacy journey, dashboard and access history help users understand and control their actual configuration.

The next step is to create low-fidelity sketches and test whether users can correctly interpret the information, whether the colour levels feel trustworthy, and which concepts provide the most value without creating excessive cognitive load.

# 6. Obstacle

## 6.1 Technical Obstacle

Different CCTV brands and configurations use different combinations of cloud storage, local storage, peer-to-peer connections, encryption, subscriptions and third-party services. Therefore, the project cannot assume that all home CCTV systems process data in the same way.

This variation may make it difficult to create one universal label or data-flow model. We will initially test the model with a small number of representative storage and access configurations. The interface should also show when information varies by model, subscription or user settings.

## 6.2 Balancing transparency and cognitive load

Providing too little information prevents users from understanding what happens to their CCTV data. However, providing too much technical information can overwhelm users and recreate the problems of traditional privacy policies.

Research on IoT security labels found that additional information did not always improve decision accuracy and could increase decision time and cognitive workload. Participants tended to prefer less cognitively demanding designs after using the labels to compare products(Caven et al., 2024).

We will use progressive disclosure: essential information will appear first, while technical details and evidence will remain available in a secondary layer.

## 6.3 Communicating access without damaging Trust

A provider may have the technical capability or contractual permission to access data without having actually viewed a user's recording. If the interface does not explain this distinction, transparency features may unnecessarily frighten users or unfairly damage trust in a provider.

The interface will distinguish access capability, authorised access, automated processing and confirmed human access. Uncertain information will be labelled as uncertain rather than presented as fact.

## 6.4 Availability and verification of manufacturer information

Manufacturers may not publicly disclose all details about backend infrastructure, staff access or third-party processing. Public information may also be outdated or difficult to verify.

Information should be cross-checked using official documentation, privacy policies, government material and credible independent research. Each label should show its source and last-updated date, and missing information should be explicitly identified.

## 6.5 User diversity

Users have different levels of technical knowledge, privacy awareness and motivation. A format that helps a technically experienced camera owner may not work for a family member who did not purchase or install the device.

Future testing should include account owners, shared household users and potential buyers, rather than only technically confident device owners.

# 7. Ethical Considerations

## 7.1 Privacy and data minimisation

CCTV footage may reveal highly private information about users, children, family members, visitors and everyday household routines. An access-history function could also create additional personal data.

The system should collect only the information necessary to explain access events. It should not create copies of CCTV footage or collect unrelated behavioural information. Access records should have a clear retention period and deletion option.

## 7.2 Accuracy and truthful communication

An inaccurate message stating that a manufacturer accessed a camera could unfairly damage trust and create unnecessary anxiety.

The design must distinguish permission from actual access and automated processing from human viewing. It should not use absolute claims such as “completely secure,”.

## 7.3 Security of access information

Access History may reveal camera locations, account activity, connected devices and patterns indicating when household members are at home.

Sensitive access information should only be visible to authorised users. Strong authentication, role-based permissions and limited data retention should be considered.

## 7.4 Consent and people who do not own the camera

Visitors, neighbours and family members may be recorded even though they did not purchase the camera, create the account or accept the provider’s privacy policy.

The project should not treat the account owner as the only stakeholder. Future research should investigate how affected people can be informed and what forms of notice or household consent are appropriate. Camera placement and recording-boundary guidance may also be included.

## 7.5 Household power imbalance

A detailed dashboard could allow the account owner to monitor how other household members use the system. This could strengthen unequal control within a household.

The system should avoid turning transparency into interpersonal surveillance. Different user roles should be clearly explained, and shared users should be informed about what activity is visible to the account owner.

## 7.6 Accessibility and colour use

A colour-coded label may be difficult for colour-blind users and may also communicate risk

Colour should always be combined with text, icons and level names. The design should not rely on red and green alone to communicate risk.

# 8. Project Risks

| Risk | Possible impact | Mitigation |
|---|---|---|

| | | |
|---|---|---|
| Difficulty recruiting enough CCTV users | Research findings may represent only a narrow user group | Recruit current users, shared household users and potential buyers through personal networks; clearly report sample limitations |
| Manufacturer information is incomplete | Labels or data-flow models may be inaccurate | Compare official documentation, privacy policies, government sources and academic research; mark unverified information as unknown |
| The problem scope becomes too broad | The project may attempt to solve security, privacy, consent and surveillance simultaneously | Focus on helping home CCTV users understand data access and handling, while documenting related issues as future research |
| Selecting a solution too early | The team may develop a polished interface without sufficient evidence | Create and test several low-fidelity concepts before selecting or combining a direction |
| Privacy risk ratings are misunderstood | Users may treat the label as a guarantee that a product is safe or unsafe | Publish transparent assessment criteria, show evidence and uncertainty, and avoid absolute claims |
| Too much information increases cognitive load | The proposed solution may reproduce the problems of existing privacy policies | Use progressive disclosure and test comprehension, decision time and perceived workload |
| Prototype testing produces socially desirable answers | Participants may say privacy is important without using the features in practice | Use task-based testing and observe what participants notice, select and understand |
| Limited semester time | The team may attempt to prototype too many connected features | Prioritise the concept that addresses the strongest research finding and use the other concepts for comparison |

Existing solutions provide fragmented privacy information. Therefore, we are exploring several connected but still provisional concepts that make CCTV data practices visible before purchase, during setup and throughout everyday use.

# 9. Project Plan

**Week 5 → Research Consolidation**
Review background research, stakeholder analysis, interviews, mental models and affinity mapping. Finalise the key findings and design requirements.
**Week 6 → Low-Fidelity Prototyping**
Develop low-fidelity sketches and mock-ups based on the initial concepts, including the Privacy Label, Privacy Journey, Privacy Dashboard and Access History.
**Week 7 → Prototype Testing**
Test the low-fidelity concepts with users. Focus on whether users can understand data storage, data flow, manufacturer access and privacy information.
**Week 8 → Analysis & Refinement**
Analyse user feedback and compare the concepts. Identify confusing information and decide which concepts or features should be kept, changed or combined.
**Week 9–10 → High-Fidelity Prototyping**
Develop the selected design direction into a higher-fidelity interactive prototype.
**Week 11 → Final User Evaluation**

Evaluate the final prototype with users. Test usability, understanding, transparency and cognitive load.
**Week 12 → Final Refinement & Documentation**
Refine the final design based on evaluation results. Complete the final proposal, design rationale and project documentation.

# 10. Appendix - AI Use Statement

| Tool | Use | Prompts | How AI suggestions were reviewed/modified | Section | Date |
|---|---|---|---|---|---|
| **Google translation** | Translation | | The team compared the translation with the original Chinese content and manually corrected wording to ensure the original meaning and participant responses were preserved. | Entire work | 23/8/2026 |
| **ChatGPT** | Organising interview findings | Helping group interview responses into possible themes and summarising repeated patterns across participants. | The team checked all suggested themes against the original interview transcripts. Themes or interpretations that were not supported by participant responses were removed or revised. | User Research | 23/8/2026 |
| **ChatGPT** | Stakeholder Map checking | Checking whether the identified stakeholders and relationships in the stakeholder map were reasonable and clearly represented. | The team compared AI suggestions with our own research and project context. Only relationships supported by our research were retained, and the final stakeholder map was manually revised by the team. | Stakeholders | 24/8/2026 |
| **ChatGPT** | Grammar and clarity checking | Checking grammar, sentence structure and clarity in report drafts. | The team reviewed suggested edits and only accepted changes that preserved the original meaning. Technical terms, research findings and claims were checked against the original sources. | Entire work | 24/8/2026 |

| | | | | | |
|---|---|---|---|---|---|
| **ChatGPT** | Report structure and organisation | Suggesting ways to organise sections and improve the logical flow between background research, user research, findings and design concepts. | The team compared the suggestions with the assessment requirements and reorganised the report manually. AI suggestions that did not match the project or assessment brief were not used. | Entire report | 25/8/2026 |
| **ChatGPT** | CCTV brand comparison support | Assisting with identifying suitable comparison categories for common home security camera brands, including local/cloud storage, remote access, encryption and data access, and helping organise the findings into a comparison table. | The team verified product features using manufacturers' official documentation and privacy policies. AI-generated descriptions were revised where features differed by model, configuration or subscription, and unsupported assumptions were excluded from the final comparison. | Problem Validation / Brand Comparison | 22/8/2026 |
| **ChatGPT** | Evidence and claim checking | Checking whether research findings, government sources and documented CCTV incidents appropriately supported the claims made in the problem validation section, and identifying claims that might be too broad or require further investigation. | The team reviewed the original sources and compared them with the suggested interpretations. Claims were narrowed or rewritten where the available evidence did not directly support them, and final conclusions were determined by the team based on the original sources. | Problem Identification / Problem Validation | 25/8/2026 |

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